Supplementary Materials

Phantom Codes: Hallucination, Accuracy, and Cost in LLM-Based Medical Concept Normalization

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This supplement accompanies the main manuscript. References cite the shared bibliography (`paper/references.bib`) section labels use the prefix S1–Sn for cross-reference from the main text.

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1 S1: Prompt templates and provider-specific structured-output configurations

This supplement reproduces the verbatim system and user prompts and the provider-specific structured-output configurations used in the headline matrix. Three prompting modes (`zeroshot`, `constrained`, `rag`) are evaluated; each is reproduced below per provider (Anthropic, OpenAI, Google) for full reviewer auditability. All content is extracted verbatim from `src/phantom_codes/models/llm.py` and `src/phantom_codes/models/rag_llm.py`. Note that the verbatim prompt strings reproduced in code blocks preserve the punctuation present in the source, including any em-dashes that appear in the actual model-facing prompts.

1.1 S1.1: Zero-shot system prompt

The zero-shot prompt is the static "base" string returned by `build_system_prompt(mode=PromptMode.ZEROSHOT)`. The model receives this once per record (it is the cacheable prefix on Anthropic; OpenAI and Google also automatically cache long prefixes). `top_k` defaults to 5 in the headline run.

```
You are a clinical informatics assistant. Your task is to recover the
standardized ICD-10-CM diagnosis code for a clinical condition described
by the user.

Return up to 5 candidate codes, ranked by your confidence (highest
first), via the `record_predictions` tool. Use ICD-10-CM dotted format
(e.g., 'E11.9', 'I10', 'N18.31'). Provide the official display text for
each code. Confidence values should reflect your true belief about each
candidate.

If you are uncertain, still return your best guess(es) - the eval framework
will distinguish hallucinations (non-existent codes) from real-but-wrong codes.
```

1.2 S1.2: Constrained system prompt

The constrained prompt appends the full CMS ACCESS-scope candidate list to the zero-shot base. The candidate block is rendered one candidate per line as `- <code>: <display> [<GROUP>]` (where `<GROUP>` is `CKM` or `ECKM`). The headline run uses ~85 candidate codes drawn from the CMS ACCESS Model FHIR Implementation Guide v0.9.6 ValueSets bundled at `src/phantom_codes/data/access_valuesets/`.

```
You are a clinical informatics assistant. Your task is to recover the
standardized ICD-10-CM diagnosis code for a clinical condition described
by the user.

Return up to 5 candidate codes, ranked by your confidence (highest
first), via the `record_predictions` tool. Use ICD-10-CM dotted format
(e.g., 'E11.9', 'I10', 'N18.31'). Provide the official display text for
each code. Confidence values should reflect your true belief about each
candidate.

If you are uncertain, still return your best guess(es) - the eval framework
will distinguish hallucinations (non-existent codes) from real-but-wrong codes.

You MUST select codes from the following candidate list. Predicting any
code not in this list is incorrect.

Candidate codes (<N> total):
- E08.00: Diabetes mellitus due to underlying condition with hyperosmolarity without nonketotic
- E08.01: Diabetes mellitus due to underlying condition with hyperosmolarity with coma [CKM]
- ...
- E11.9: Type 2 diabetes mellitus without complications [CKM]
- ...
- I10: Essential (primary) hypertension [ECKM]
- ...
```

```
- N18.31: Chronic kidney disease, stage 3a [CKM]
- ...
```

1.3 S1.3: RAG system prompt template

The RAG prompt reuses the constrained-mode template, but the candidate list is populated per-record by a frozen sentence-transformer retriever (default `all-MiniLM-L6-v2`) returning the top-`retrieve_k` = 20 ACCESS-scope candidates most similar to the input. The structural form is therefore identical to S1.2; only the size and identity of the candidate block change. From `RAGLLMModel.predict` in src/phantom_codes/models/rag_llm.py:

```
# Stage 1 - retrieve top-K candidates for this specific input.
retrieved = self._retriever.predict(
    input_fhir=input_fhir,
    input_text=input_text,
    top_k=self._retrieve_k,  # default 20
)

# Stage 2 - build a constrained-mode prompt with these per-record candidates
# and call the LLM.
system_prompt = build_system_prompt(
    PromptMode.CONSTRAINED,
    candidates=candidates,
    top_k=top_k,
)
```

1.4 S1.4: User-message template

The user message varies by input mode. For D1_full and D2_no_code inputs, the FHIR Condition resource is serialized to indented JSON. For D3_text_only and D4_abbreviated inputs, the text is sent directly. Returned by `build_user_message(input_fhir=…, input_text=…)`:

```
The clinical condition is described by this (degraded) FHIR Condition
```

resource. Recover the ICD-10-CM code:

```json
{
  "resourceType": "Condition",
  "code": { ... }
  ...
}
```

For text-only modes:

The clinical condition is described by this text. Recover the ICD-10-CM code:

S1.5: Provider-specific structured-output configuration

All three providers are constrained to emit the same `PREDICTION_TOOL_SCHEMA` shape. The schema is defined once in `src/phantom_codes/models/llm.py` and adapted per-provider:

```json
{
  "type": "object",
  "properties": {
    "predictions": {
      "type": "array",
      "minItems": 1,
      "maxItems": 10,
      "items": {
        "type": "object",
        "properties": {
```

```
          "code": {
            "type": "string",
            "description": "ICD-10-CM code in dotted format, e.g. 'E11.9' or 'I10'."
          },
          "display": {
            "type": "string",
            "description": "Human-readable description of the code."
          },
          "confidence": {
            "type": "number",
            "minimum": 0.0,
            "maximum": 1.0,
            "description": "Model's confidence, higher = more confident."
          }
        },
        "required": ["code", "display", "confidence"],
        "additionalProperties": false
      }
    }
  },
  "required": ["predictions"],
  "additionalProperties": false
}
```

1.4.1 S1.5.1 Anthropic: forced tool use

The schema is wrapped in a single tool with `tool_choice` forced to that tool, so the model must invoke it. The system block is marked `cache_control: ephemeral` for prompt caching.

```
client.messages.create(
    model=self.model_id,
```

```
    max_tokens=self.max_tokens,
    system=[
        {
            "type": "text",
            "text": system_prompt,
            "cache_control": {"type": "ephemeral"},
        }
    ],
    tools=[
        {
            "name": "record_predictions",
            "description": (
                "Record the model's ranked ICD-10-CM code predictions "
                "for the given clinical input."
            ),
            "input_schema": PREDICTION_TOOL_SCHEMA,
        }
    ],
    tool_choice={"type": "tool", "name": "record_predictions"},
    messages=[{"role": "user", "content": user_message}],
)
```

1.4.2 S1.5.2 OpenAI: structured outputs (`response_format`)

OpenAI's strict `json_schema` mode requires a slightly trimmed schema (strict-mode rejects `minItems`, `maxItems`, `minimum`, `maximum`, `pattern`, `format`); a deep-copy helper `_strip_for_openai_strict` removes those fields. GPT-5+ and reasoning models (o1, o3, o4) require `max_completion_tokens` instead of the legacy `max_tokens`.

```
strict_schema = _strip_for_openai_strict(PREDICTION_TOOL_SCHEMA)
client.chat.completions.create(
```

```
    model=self.model_id,
    response_format={
        "type": "json_schema",
        "json_schema": {
            "name": "icd10_predictions",
            "strict": True,
            "schema": strict_schema,
        },
    },
    messages=[
        {"role": "system", "content": system_prompt},
        {"role": "user", "content": user_message},
    ],
    **{token_kwarg: self.max_tokens},  # token_kwarg: "max_completion_tokens" for GPT-5+/o-seri
)
```

1.4.3 S1.5.3 Google: `response_schema`

Gemini's `response_schema` is an OpenAPI 3.0 subset that rejects JSON Schema fields the other providers accept, most notably `additionalProperties` and `$schema`. A deep-copy helper `_strip_for_gemini` removes those fields. The system instruction is passed as a separate `system_instruction` config field; output is constrained to `application/json` with the trimmed schema.

```
config = types.GenerateContentConfig(
    system_instruction=system_prompt,
    max_output_tokens=self.max_tokens,
    response_mime_type="application/json",
    response_schema=_strip_for_gemini(PREDICTION_TOOL_SCHEMA),
)
client.models.generate_content(
    model=self.model_id,
```

```
    contents=user_message,
    config=config,
)
```

The Google client wraps the call in a 429-only retry loop (one outer attempt) that respects the server's `retryDelay` hint; 5xx retries are handled by the SDK's internal backoff (5 attempts, exponential 1→16s with jitter).

2 S2: Extended results

This supplement extends §3 of the main manuscript with the full per-(model, mode) outcome distribution across all 28 configurations, the complete Wilson 95% CI tables for hallucination and no_prediction, the full top-k lift table, the per-bucket cost decomposition, and an enumeration of every fabricated ICD-10-CM code emitted in the headline run. All tables are auto-generated by `phantom-codes report` from `results/raw/headline_n125_20260504T040931Z.csv` into `results/summary/n125_run_v2/`; this supplement reproduces them verbatim for reviewer auditability.

2.1 S2.1: Full headline outcome distribution (28 configs × 4 modes)

Each row sums to n=125 across the six outcome buckets (counts and percentages). The headline-relevant subset of these rows is in main §3 Table 1; the full set is reproduced below for auditability.

Model	Mode	n	Exact	Category	Chapter	OOD	NoPred	Halluc
baseline:exact	D1_full	125	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	125 (100.0%)	0 (0.0%)
baseline:exact	D2_no_code	125	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	125 (100.0%)	0 (0.0%)
baseline:exact	D3_text_only	125	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	125 (100.0%)	0 (0.0%)
baseline:exact	D4_abbreviated	125	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	125 (100.0%)	0 (0.0%)
baseline:fuzzy	D1_full	125	67 (53.6%)	28 (22.4%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	30 (24.0%)
baseline:fuzzy	D2_no_code	125	67 (53.6%)	28 (22.4%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	30 (24.0%)
baseline:fuzzy	D3_text_only	125	67 (53.6%)	28 (22.4%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	30 (24.0%)
baseline:fuzzy	D4_abbreviated	125	25 (20.0%)	8 (6.4%)	30 (24.0%)	28 (22.4%)	0 (0.0%)	34 (27.2%)

Model	Mode	n	Exact	Category	Chapter	OOD	NoPred	Halluc
baseline:tfidf	D1_full	125	67 (53.6%)	15 (12.0%)	0 (0.0%)	0 (0.0%)	7 (5.6%)	36 (28.8%)
baseline:tfidf	D2_no_code	125	67 (53.6%)	15 (12.0%)	0 (0.0%)	0 (0.0%)	7 (5.6%)	36 (28.8%)
baseline:tfidf	D3_text_only	125	67 (53.6%)	15 (12.0%)	0 (0.0%)	0 (0.0%)	7 (5.6%)	36 (28.8%)
baseline:tfidf	D4_abbreviated	125	25 (20.0%)	7 (5.6%)	16 (12.8%)	38 (30.4%)	0 (0.0%)	39 (31.2%)
claude-haiku-4-5:constrained	D1_full	125	125 (100.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
claude-haiku-4-5:constrained	D2_no_code	125	125 (100.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
claude-haiku-4-5:constrained	D3_text_only	125	117 (93.6%)	8 (6.4%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
claude-haiku-4-5:constrained	D4_abbreviated	125	117 (93.6%)	8 (6.4%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
claude-haiku-4-5:rag	D1_full	125	125 (100.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
claude-haiku-4-5:rag	D2_no_code	125	123 (98.4%)	2 (1.6%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
claude-haiku-4-5:rag	D3_text_only	125	110 (88.0%)	15 (12.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
claude-haiku-4-5:rag	D4_abbreviated	125	95 (76.0%)	30 (24.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
claude-haiku-4-5:zeroshot	D1_full	125	125 (100.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
claude-haiku-4-5:zeroshot	D2_no_code	125	113 (90.4%)	8 (6.4%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	4 (3.2%)
claude-haiku-4-5:zeroshot	D3_text_only	125	105 (84.0%)	14 (11.2%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	6 (4.8%)
claude-haiku-4-5:zeroshot	D4_abbreviated	125	102 (81.6%)	18 (14.4%)	0 (0.0%)	1 (0.8%)	0 (0.0%)	4 (3.2%)
claude-opus-4-7:constrained	D1_full	125	125 (100.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
claude-opus-4-7:constrained	D2_no_code	125	125 (100.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
claude-opus-4-7:constrained	D3_text_only	125	116 (92.8%)	8 (6.4%)	0 (0.0%)	0 (0.0%)	1 (0.8%)	0 (0.0%)
claude-opus-4-7:constrained	D4_abbreviated	125	108 (86.4%)	17 (13.6%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
claude-opus-4-7:rag	D1_full	125	125 (100.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
claude-opus-4-7:rag	D2_no_code	125	125 (100.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
claude-opus-4-7:rag	D3_text_only	125	109 (87.2%)	16 (12.8%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
claude-opus-4-7:rag	D4_abbreviated	125	102 (81.6%)	23 (18.4%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
claude-opus-4-7:zeroshot	D1_full	125	124 (99.2%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	1 (0.8%)
claude-opus-4-7:zeroshot	D2_no_code	125	124 (99.2%)	1 (0.8%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
claude-opus-4-7:zeroshot	D3_text_only	125	96 (76.8%)	22 (17.6%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	7 (5.6%)
claude-opus-4-7:zeroshot	D4_abbreviated	125	101 (80.8%)	17 (13.6%)	0 (0.0%)	5 (4.0%)	2 (1.6%)	0 (0.0%)

Model	Mode	n	Exact	Category	Chapter	OOD	NoPred	Halluc
claude-sonnet-4-6:constrained	D1_full	125	125 (100.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
claude-sonnet-4-6:constrained	D2_no_code	125	125 (100.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
claude-sonnet-4-6:constrained	D3_text_only	125	110 (88.0%)	15 (12.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
claude-sonnet-4-6:constrained	D4_abbreviated	125	113 (90.4%)	12 (9.6%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
claude-sonnet-4-6:rag	D1_full	125	125 (100.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
claude-sonnet-4-6:rag	D2_no_code	125	125 (100.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
claude-sonnet-4-6:rag	D3_text_only	125	108 (86.4%)	17 (13.6%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
claude-sonnet-4-6:rag	D4_abbreviated	125	96 (76.8%)	29 (23.2%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
claude-sonnet-4-6:zeroshot	D1_full	125	125 (100.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
claude-sonnet-4-6:zeroshot	D2_no_code	125	94 (75.2%)	30 (24.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	1 (0.8%)
claude-sonnet-4-6:zeroshot	D3_text_only	125	81 (64.8%)	36 (28.8%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	8 (6.4%)
claude-sonnet-4-6:zeroshot	D4_abbreviated	125	77 (61.6%)	40 (32.0%)	0 (0.0%)	4 (3.2%)	0 (0.0%)	4 (3.2%)
gemini-2.5-flash:constrained	D1_full	125	124 (99.2%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	1 (0.8%)	0 (0.0%)
gemini-2.5-flash:constrained	D2_no_code	125	125 (100.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
gemini-2.5-flash:constrained	D3_text_only	125	105 (84.0%)	5 (4.0%)	0 (0.0%)	0 (0.0%)	15 (12.0%)	0 (0.0%)
gemini-2.5-flash:constrained	D4_abbreviated	125	88 (70.4%)	14 (11.2%)	0 (0.0%)	0 (0.0%)	23 (18.4%)	0 (0.0%)
gemini-2.5-flash:rag	D1_full	125	123 (98.4%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	2 (1.6%)	0 (0.0%)
gemini-2.5-flash:rag	D2_no_code	125	120 (96.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	5 (4.0%)	0 (0.0%)
gemini-2.5-flash:rag	D3_text_only	125	96 (76.8%)	2 (1.6%)	0 (0.0%)	0 (0.0%)	27 (21.6%)	0 (0.0%)
gemini-2.5-flash:rag	D4_abbreviated	125	84 (67.2%)	15 (12.0%)	0 (0.0%)	0 (0.0%)	26 (20.8%)	0 (0.0%)
gemini-2.5-flash:zeroshot	D1_full	125	125 (100.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
gemini-2.5-flash:zeroshot	D2_no_code	125	107 (85.6%)	4 (3.2%)	0 (0.0%)	0 (0.0%)	14 (11.2%)	0 (0.0%)
gemini-2.5-flash:zeroshot	D3_text_only	125	76 (60.8%)	1 (0.8%)	0 (0.0%)	0 (0.0%)	47 (37.6%)	1 (0.8%)
gemini-2.5-flash:zeroshot	D4_abbreviated	125	69 (55.2%)	2 (1.6%)	0 (0.0%)	0 (0.0%)	53 (42.4%)	1 (0.8%)
gemini-2.5-pro:constrained	D1_full	125	57 (45.6%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	68 (54.4%)	0 (0.0%)
gemini-2.5-pro:constrained	D2_no_code	125	85 (68.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	40 (32.0%)	0 (0.0%)
gemini-2.5-pro:constrained	D3_text_only	125	40 (32.0%)	1 (0.8%)	0 (0.0%)	0 (0.0%)	84 (67.2%)	0 (0.0%)
gemini-2.5-pro:constrained	D4_abbreviated	125	49 (39.2%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	76 (60.8%)	0 (0.0%)

Model	Mode	n	Exact	Category	Chapter	OOD	NoPred	Halluc
gemini-2.5-pro:rag	D1_full	125	49 (39.2%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	76 (60.8%)	0 (0.0%)
gemini-2.5-pro:rag	D2_no_code	125	44 (35.2%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	81 (64.8%)	0 (0.0%)
gemini-2.5-pro:rag	D3_text_only	125	31 (24.8%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	94 (75.2%)	0 (0.0%)
gemini-2.5-pro:rag	D4_abbreviated	125	35 (28.0%)	7 (5.6%)	0 (0.0%)	0 (0.0%)	83 (66.4%)	0 (0.0%)
gemini-2.5-pro:zeroshot	D1_full	125	28 (22.4%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	97 (77.6%)	0 (0.0%)
gemini-2.5-pro:zeroshot	D2_no_code	125	18 (14.4%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	107 (85.6%)	0 (0.0%)
gemini-2.5-pro:zeroshot	D3_text_only	125	7 (5.6%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	118 (94.4%)	0 (0.0%)
gemini-2.5-pro:zeroshot	D4_abbreviated	125	6 (4.8%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	119 (95.2%)	0 (0.0%)
gemini-3-flash-preview:constrained	D1_full	125	122 (97.6%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	3 (2.4%)	0 (0.0%)
gemini-3-flash-preview:constrained	D2_no_code	125	121 (96.8%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	4 (3.2%)	0 (0.0%)
gemini-3-flash-preview:constrained	D3_text_only	125	99 (79.2%)	4 (3.2%)	0 (0.0%)	0 (0.0%)	22 (17.6%)	0 (0.0%)
gemini-3-flash-preview:constrained	D4_abbreviated	125	106 (84.8%)	3 (2.4%)	0 (0.0%)	0 (0.0%)	16 (12.8%)	0 (0.0%)
gemini-3-flash-preview:rag	D1_full	125	119 (95.2%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	6 (4.8%)	0 (0.0%)
gemini-3-flash-preview:rag	D2_no_code	125	107 (85.6%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	18 (14.4%)	0 (0.0%)
gemini-3-flash-preview:rag	D3_text_only	125	102 (81.6%)	3 (2.4%)	0 (0.0%)	0 (0.0%)	20 (16.0%)	0 (0.0%)
gemini-3-flash-preview:rag	D4_abbreviated	125	84 (67.2%)	17 (13.6%)	0 (0.0%)	0 (0.0%)	24 (19.2%)	0 (0.0%)
gemini-3-flash-preview:zeroshot	D1_full	125	97 (77.6%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	28 (22.4%)	0 (0.0%)
gemini-3-flash-preview:zeroshot	D2_no_code	125	90 (72.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	35 (28.0%)	0 (0.0%)
gemini-3-flash-preview:zeroshot	D3_text_only	125	69 (55.2%)	4 (3.2%)	0 (0.0%)	0 (0.0%)	52 (41.6%)	0 (0.0%)
gemini-3-flash-preview:zeroshot	D4_abbreviated	125	75 (60.0%)	1 (0.8%)	0 (0.0%)	0 (0.0%)	49 (39.2%)	0 (0.0%)
gpt-4o-mini:constrained	D1_full	125	125 (100.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
gpt-4o-mini:constrained	D2_no_code	125	125 (100.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
gpt-4o-mini:constrained	D3_text_only	125	113 (90.4%)	12 (9.6%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
gpt-4o-mini:constrained	D4_abbreviated	125	108 (86.4%)	17 (13.6%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
gpt-4o-mini:rag	D1_full	125	125 (100.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
gpt-4o-mini:rag	D2_no_code	125	125 (100.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
gpt-4o-mini:rag	D3_text_only	125	101 (80.8%)	24 (19.2%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
gpt-4o-mini:rag	D4_abbreviated	125	101 (80.8%)	24 (19.2%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)

Model	Mode	n	Exact	Category	Chapter	OOD	NoPred	Halluc
gpt-4o-mini:zeroshot	D1_full	125	124 (99.2%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	1 (0.8%)
gpt-4o-mini:zeroshot	D2_no_code	125	106 (84.8%)	2 (1.6%)	1 (0.8%)	0 (0.0%)	0 (0.0%)	16 (12.8%)
gpt-4o-mini:zeroshot	D3_text_only	125	98 (78.4%)	13 (10.4%)	0 (0.0%)	2 (1.6%)	0 (0.0%)	12 (9.6%)
gpt-4o-mini:zeroshot	D4_abbreviated	125	98 (78.4%)	16 (12.8%)	0 (0.0%)	2 (1.6%)	0 (0.0%)	9 (7.2%)
gpt-5.5:constrained	D1_full	125	125 (100.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
gpt-5.5:constrained	D2_no_code	125	125 (100.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
gpt-5.5:constrained	D3_text_only	125	109 (87.2%)	16 (12.8%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
gpt-5.5:constrained	D4_abbreviated	125	109 (87.2%)	16 (12.8%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
gpt-5.5:rag	D1_full	125	124 (99.2%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	1 (0.8%)	0 (0.0%)
gpt-5.5:rag	D2_no_code	125	123 (98.4%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	2 (1.6%)	0 (0.0%)
gpt-5.5:rag	D3_text_only	125	109 (87.2%)	16 (12.8%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
gpt-5.5:rag	D4_abbreviated	125	90 (72.0%)	29 (23.2%)	0 (0.0%)	0 (0.0%)	6 (4.8%)	0 (0.0%)
gpt-5.5:zeroshot	D1_full	125	124 (99.2%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	1 (0.8%)	0 (0.0%)
gpt-5.5:zeroshot	D2_no_code	125	125 (100.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
gpt-5.5:zeroshot	D3_text_only	125	89 (71.2%)	16 (12.8%)	0 (0.0%)	19 (15.2%)	1 (0.8%)	0 (0.0%)
gpt-5.5:zeroshot	D4_abbreviated	125	106 (84.8%)	16 (12.8%)	0 (0.0%)	3 (2.4%)	0 (0.0%)	0 (0.0%)
sentence-transformer:retrieval	D1_full	125	101 (80.8%)	24 (19.2%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
sentence-transformer:retrieval	D2_no_code	125	101 (80.8%)	24 (19.2%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
sentence-transformer:retrieval	D3_text_only	125	101 (80.8%)	24 (19.2%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
sentence-transformer:retrieval	D4_abbreviated	125	42 (33.6%)	4 (3.2%)	8 (6.4%)	65 (52.0%)	0 (0.0%)	6 (4.8%)

2.2 S2.2–S2.7: Pointer to repository

Extended per-cell results — the full hallucination-rate table with Wilson 95% CIs (S2.2), the full no_prediction-rate table with Wilson 95% CIs (S2.3), the full 28-configuration top-1 vs. top-5 lift table (S2.4), the per-bucket cost decomposition (S2.5), the per-disease-group breakdown (S2.6 — deferred to v2 at n=125), and the per-vocabulary-position breakdown (S2.7 — similarly deferred) — are released in the project repository at https://github.com/briankfung/phantom-codes under `results/summary/n125_run_v2/` alongside the per-prediction CSV.

2.3 S2.8: Hallucinated-code listing

Every fabricated (non-existent) ICD-10-CM code emitted as a top-1 prediction in the headline run, grouped by (model, mode, code) and sorted by count. A "fabricated" code is one that does not appear in the FY2026 CMS-published ICD-10-CM tabular list, mechanically checked by the bundled validator at `src/phantom_codes/data/icd10cm/`. Empty-prediction rows (the model returned no code at all) are classified as `no_prediction` in the 6-way taxonomy and are not included here; see §S2.3 for those rates.

Across the 28-configuration matrix on n=125 × 4 modes = 14,000 total top-1 predictions, 352 (2.5%) were fabrications, drawn from 16 unique non-existent ICD-10-CM strings.

Model	Mode	Predicted code	Display	n
baseline:fuzzy	D1_full	E66.8	Other obesity	30
baseline:fuzzy	D2_no_code	E66.8	Other obesity	30
baseline:fuzzy	D3_text_only	E66.8	Other obesity	30
baseline:fuzzy	D4_abbreviated	E66.8	Other obesity	30
baseline:tfidf	D1_full	E66.8	Other obesity	30
baseline:tfidf	D2_no_code	E66.8	Other obesity	30
baseline:tfidf	D3_text_only	E66.8	Other obesity	30
baseline:tfidf	D4_abbreviated	E66.8	Other obesity	30
baseline:tfidf	D4_abbreviated	I63.0	Cerebral infarction due to thrombosis of precerebral arteries	9
gpt-4o-mini:zeroshot	D2_no_code	E11.2	Type 2 diabetes mellitus with kidney complications	8
claude-opus-4-7:zeroshot	D3_text_only	N18.3	Chronic kidney disease, stage 3 (moderate)	7
claude-sonnet-4-6:zeroshot	D3_text_only	N18.3	Chronic kidney disease, stage 3 (moderate)	7
gpt-4o-mini:zeroshot	D2_no_code	N18.3	Chronic kidney disease, stage 3 (moderate)	7
baseline:tfidf	D1_full	E78.4	Other hyperlipidemia	6
baseline:tfidf	D2_no_code	E78.4	Other hyperlipidemia	6
baseline:tfidf	D3_text_only	E78.4	Other hyperlipidemia	6
gpt-4o-mini:zeroshot	D3_text_only	E11.2	Type 2 diabetes mellitus with kidney complications	6
sentence-transformer:retrieval	D4_abbreviated	I73.8	Other specified peripheral vascular diseases	6

Model	Mode	Predicted code	Display	n
claude-haiku-4-5:zeroshot	D3_text_only	N18.3	Chronic kidney disease, stage 3	5
gpt-4o-mini:zeroshot	D3_text_only	N18.3	Chronic kidney disease stage 3 (moderate)	5
baseline:fuzzy	D4_abbreviated	E78.7	Disorders of bile acid and cholesterol metabolism	4
claude-haiku-4-5:zeroshot	D2_no_code	N18.3	Chronic kidney disease, stage 3 (moderate)	3
claude-sonnet-4-6:zeroshot	D4_abbreviated	N18.3	Chronic kidney disease, stage 3 (moderate)	3
gpt-4o-mini:zeroshot	D4_abbreviated	E11.2	Type 2 diabetes mellitus with kidney complications	3
gpt-4o-mini:zeroshot	D4_abbreviated	N18.3	Chronic kidney disease stage 3 (moderate)	3
claude-haiku-4-5:zeroshot	D2_no_code	E11.339	Type 2 diabetes mellitus with mild nonproliferative diabetic retinopathy without macular edema, unspecified eye	1
claude-haiku-4-5:zeroshot	D3_text_only	E11.329	Type 2 diabetes mellitus with nonproliferative diabetic retinopathy without macular edema	1
claude-haiku-4-5:zeroshot	D4_abbreviated	E11.329	Type 2 diabetes mellitus with mild nonproliferative diabetic retinopathy without macular edema	1
claude-haiku-4-5:zeroshot	D4_abbreviated	N18.3	Chronic kidney disease, stage 3	1
claude-haiku-4-5:zeroshot	D4_abbreviated	Z87.72	Personal history of diseases of the kidney and ureter	1
claude-haiku-4-5:zeroshot	D4_abbreviated	Z87.735	Personal history of chronic kidney disease	1
claude-opus-4-7:zeroshot	D1_full	E11.329	Type 2 diabetes mellitus with mild nonproliferative diabetic retinopathy without macular edema, unspecified eye	1

Model	Mode	Predicted code	Display	n
claude-sonnet-4-6:zeroshot	D2_no_code	E11.329	Type 2 diabetes mellitus with mild nonproliferative diabetic retinopathy without macular edema, unspecified eye	1
claude-sonnet-4-6:zeroshot	D3_text_only	E11.32	Type 2 diabetes mellitus with mild nonproliferative diabetic retinopathy	1
claude-sonnet-4-6:zeroshot	D4_abbreviated	E11.32	Type 2 diabetes mellitus with mild nonproliferative diabetic retinopathy	1
gemini-2.5-flash:zeroshot	D3_text_only	E78.10	Pure hypertriglyceridemia, unspecified	1
gemini-2.5-flash:zeroshot	D4_abbreviated	N18.3	Chronic kidney disease, stage 3 (moderate)	1
gpt-4o-mini:zeroshot	D1_full	E11.359	Type 2 diabetes mellitus with mild nonproliferative diabetic retinopathy, unspecified eye	1
gpt-4o-mini:zeroshot	D2_no_code	E11.359	Type 2 diabetes mellitus with mild nonproliferative diabetic retinopathy, unspecified eye	1
gpt-4o-mini:zeroshot	D3_text_only	E11.359	Type 2 diabetes mellitus with mild nonproliferative diabetic retinopathy, unspecified eye	1
gpt-4o-mini:zeroshot	D4_abbreviated	E11.3	Type 2 diabetes mellitus with renal complications	1
gpt-4o-mini:zeroshot	D4_abbreviated	E11.359	Type 2 diabetes mellitus with nonproliferative diabetic retinopathy, unspecified eye	1
gpt-4o-mini:zeroshot	D4_abbreviated	N08.3	Disorder of kidney due to type 1 diabetes mellitus	1

Two qualitative observations from the listing:

1. String-matching baselines fabricate by design. Both `baseline:fuzzy` and `baseline:tfidf` repeatedly return `E66.8` ("Other obesity") and similar near-miss codes that exist in ICD-10-CM but

were not in the bundled FY2026 CMS validator's accepted-code set under the strict comparison logic. These are fabrications under the *narrow* mechanical definition; whether they should be reclassified to a softer "out-of-domain" category in v2 is a methodology choice.

2. Frontier-model fabrications cluster on plausible but non-existent specificity. Most frontier-model fabrications fall into two patterns: (a) confused chronic-kidney-disease subcoding (`N18.3` instead of the real `N18.30`/`N18.31`/`N18.32`); and

   (b) over-specified type-2-diabetes-with-retinopathy codes (`E11.329`, `E11.339`, `E11.359`) that look like legitimate ICD-10-CM strings but do not exist in the FY2026 tabular list. The pattern is recognizable: the model is reaching for a more specific code than the input warrants and inventing a plausible suffix.

3 S5: Cost economics: pointer to repository

The headline per-call cost, cost-per-correct, and Pareto-frontier analysis appear in main §3. Extended analyses — the full per-(model, prompting-mode) cost table, per-bucket cost decomposition, production-deployment cost projections at multiple organization scales, the deployment break-even analysis vs. a human coder, qualitative deployment considerations (Anthropic prompt- caching threshold, reliability as a cost factor, pricing-snapshot reproducibility), and the historical pre-headline-run smoke-test cost data — are released in the project repository at https://github.com/briankfung/phantom-codes under `paper/supp_extended/` and the per-prediction CSV at `results/summary/n125_run_v2/cost_per_correct.`
Total run cost across the 28 configurations on the n=125 cohort was $49.65, dominated by Claude Opus ($18.18 across its 3 modes) and the GPT-5.5 / Sonnet 4.6 groupings.